MindZed

BIOARC

Compact autonomous microalgae photobioreactor

THE PROBLEM

Carbon emissions, wastewater and land constraints are usually treated as separate problems.

We saw an opportunity to connect them through one compact biological system.

OUR HARDWARE RESPONSE

BioArc cultivates microalgae to capture CO_2 and generate O_2 through photosynthesis, while using nutrient-rich municipal wastewater as a growth medium.

The harvested biomass becomes a feedstock for biofuels, biogas and biopolymers.

CO_2 → O_2 | WASTEWATER → RESOURCE | BIOMASS → VALUE



WORKING PHYSICAL PROTOTYPE

sense • decide • act • observe

A compact biological system designed for high-output, resource-efficient deployment.

01 / THE CHALLENGE

Two gaps. One opportunity.

The environmental need is clear. The engineering challenge is making the biological process stable, measurable, and useful outside a laboratory.

01

CARBON + WASTE

CO_2 needs scalable capture, while nutrient-rich wastewater still requires treatment.

02

FOOTPRINT

Meaningful biological impact is difficult to deliver when the solution depends on large land areas.

03

BIOLOGICAL DRIFT

Light, temperature, aeration, level and growth can move out of range and destabilize a dense culture.

04

CONTROL COMPLEXITY

Without integrated sensing, actuation and telemetry, optimization becomes manual, slow and expensive.

DESIGN THESIS

Build the biology as a feedback system, not just a tank.

BioArc pairs the physical reactor with edge sensing, local control, real-time telemetry and an AI-assisted software layer.

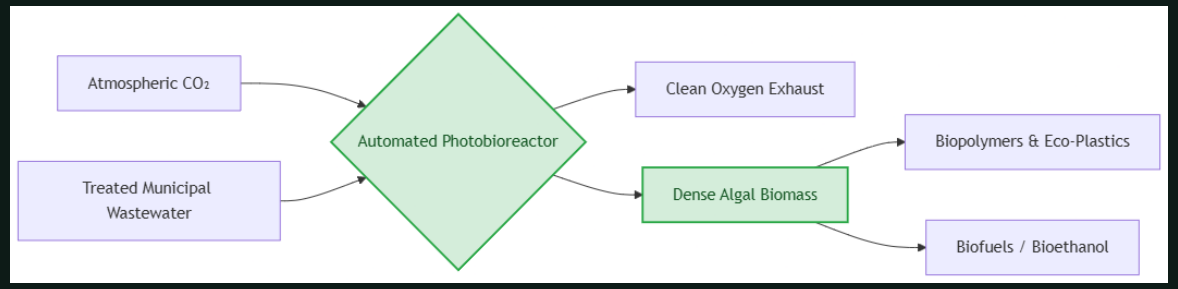
The hardware creates the biological output. The software makes that output controllable and observable.

02 / SOLUTION

One reactor. Multiple outputs.

BioArc turns a compact microalgae culture into a circular resource engine: carbon goes in, oxygen and useful resources come out.

PHYSICAL + RESOURCE LOOP



Reference lifecycle from the BioArc prototype documentation.

01 CARBON CAPTURE

Microalgae use incoming CO_2 as a carbon source during photosynthesis.

02 OXYGEN GENERATION

Photosynthesis produces oxygen, creating a clean-gas output stream.

03 WATER RECOVERY

Nutrient uptake reduces the load of the cultivation medium, followed by downstream polishing for reuse.

04 BIOMASS VALORISATION

Harvested algal residue becomes a feedstock for biofuels, biogas and biopolymers.

Carbon captured • Water recovered • Biomass valorised

AUTONOMOUS CULTURE CYCLE

START → INTAKE → GROW → MEASURE → HARVEST → RESET

03 / ARCHITECTURE

Hardware at the core. Software around it.

The time-critical control loop stays at the edge. The web and AI layers add observability, analytics and operator intelligence without owning the reactor.

BIOREACTOR
Sensors + actuators

→

ESP32 EDGE
FSM + local control

→

MQTT / WS
Real-time telemetry

→

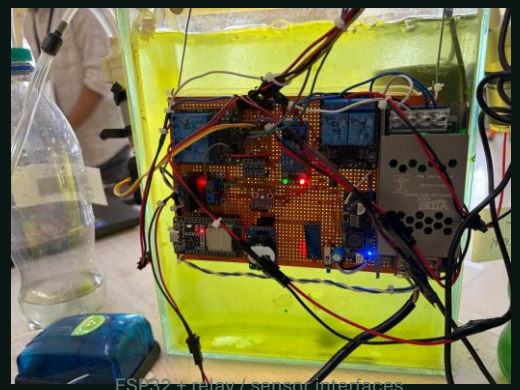
DASHBOARD
Live data + history

→

AI COPILOT
AI Control + safety-checked tools

PHYSICAL CONTROL

Sensors	Optical biomass • water temperature ambient conditions • liquid level
Actuators	Grow lights • aeration nutrient dosing • agitation • harvest



ESP32 + relay / sensor interfaces

SOFTWARE + AI LAYER

Real-time	MQTT telemetry + WebSocket streaming for live state
AI copilot	Natural-language queries + controlled actuator tools
Offline path	Local FSM + captive portal keeps field control available without cloud

04 / WHY BIOARC

Designed to be useful, controllable, and hard to break.

The differentiator is the combination of a physical biological output with a resilient sensing and software control loop.

01

COMPACT BIOLOGICAL ENGINE

High-output photosynthetic process in a footprint far smaller than conventional terrestrial greenery.

02

CIRCULAR RESOURCE USE

One system connects CO_2 capture, nutrient-rich wastewater cultivation, oxygen generation and biomass recovery.

03

EDGE AUTONOMY

Local control keeps the reactor operating during network loss; software adds visibility rather than replacing the controller.

04

LOW-COST OPTICAL INTELLIGENCE

Algorithmic biomass estimation turns inexpensive optical sensing into a practical control signal.

ENGINEERING CHALLENGES SOLVED

LED interference / bubble turbidity / network jitter / async telemetry drift

Optical normalization + quiet sampling + resilient messaging + strict data contracts

NEXT FRONTIER

Closed-loop carbon + oxygen monitoring

Add dedicated CO_2, O_2, dissolved-oxygen and pH sensing for remote or space-habitat research.

Carbon captured. Water recovered. Biomass valorised.
All inside a compact automated biological system.